AyuAware Health Section

Technical Product Requirements Document

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1. Executive Summary

What We're Building

The Health section is the **deep-dive dashboard** for understanding overall wellbeing through an Ayurvedic lens. At its center is the **CHI (Constitutional Health Index)** — a proprietary score that synthesizes multiple health dimensions into a single, actionable number.

Unlike typical health apps that show isolated metrics, the Health section presents health as an interconnected system built on **7 Ayurvedic pillars**:

Pillar	Sanskrit	Meaning	What It Measures
Prakriti	प्रकृति	Constitution	Birth constitution alignment
Vikriti	विकृति	Current State	Deviation from natural balance
Agni	अग्नि	Digestive Fire	Metabolic and digestive strength
Ojas	ओजस	Vitality	Deep reserves and immunity
Manas	मनस	Mind	Mental clarity and emotional balance
Nidra	निद्रा	Sleep	Sleep quality and patterns
Prana	प्राण	Life Force	Vital energy and breath

Why It Matters

Health data without context is overwhelming. Users see steps, sleep hours, HRV numbers — but don't know what they mean or what to do about them. The Health section solves this by:

1. **Synthesizing complexity** — CHI distills many metrics into one understandable score
2. **Providing Ayurvedic context** — Numbers become meaningful through traditional wisdom
3. **Enabling action** — Every insight connects to protocols, supplements, or lifestyle changes
4. **Teaching as it shows** — The Twin explains concepts as users explore

Key Differentiator

Most health apps are metric dumps. AyuAware's Health section is a **guided education** wrapped in a **personalized dashboard**. The Twin doesn't just show a score — it explains why that score matters for *this particular user* based on their constitution.

2. Section Overview

2.1 Information Architecture

```
Health Section
│
├── 🌟 CHI Overview
│   ├── Overall score (0-100)
│   ├── Trend comparison
│   ├── Quick insight
│   └── Detailed analysis (expanded)
│
├── 🏛️ Seven Pillars Grid
│   ├── Prakriti (Constitution) — 🌿
│   ├── Vikriti (Current State) — 🔄
│   ├── Agni (Digestive Fire) — 🔥
│   ├── Ojas (Vitality) — ✨
│   ├── Manas (Mind) — 🧠
│   ├── Nidra (Sleep) — 🌙
│   └── Prana (Life Force) — 💨
│
├── 📊 Additional Scores
│   ├── Dynamic Fluctuation Score
│   ├── Metabolic Intensity Score
│   └── System Energy Score
│
└── 🎯 Actions
```

```
    ├── Create Protocol
    ├── Shop Supplements
    └── View History
```

2.2 The CHI Score Concept

CHI (Constitutional Health Index) is the flagship metric of AyuAware. It represents:

What It Is:

- A 0-100 score synthesizing all 7 pillars
- Weighted based on user's constitution (Prakriti)
- Dynamic — changes daily based on lifestyle
- Personalized — same behaviors affect different constitutions differently

How It's Calculated:

```
CHI = Σ (Pillar Score × Pillar Weight × Constitution
Modifier)

Where:
- Pillar Score: Individual 0-100 score for each pillar
- Pillar Weight: Importance based on current health focus
- Constitution Modifier: Adjustment based on Prakriti
```

Score Ranges:

Range	Label	Color	Meaning
85-100	Excellent	#4A9B8E (green)	Thriving, maintain current practices
70-84	Good	#7AB8AC (light teal)	Solid foundation, minor optimizations
55-69	Fair	#F8C471 (gold)	Attention needed, actionable changes
0-54	Needs Attention	#E89B7F (coral)	Priority focus required

2.3 Design Philosophy

Educational First-Time Experience New users receive a gentle introduction to CHI and the 7 pillars before seeing their data. This investment in education pays dividends in engagement.

Pillar-Based Organization Rather than drowning users in metrics, we organize everything under 7 memorable pillars. Users learn to think in terms of "How's my Agni?" rather than "What's my digestive enzyme level?"

Action-Oriented Insights Every piece of data connects to something the user can do:

- Low Agni? → Create an Agni-boosting protocol
- Vata imbalance? → Shop Vata-balancing supplements
- Poor Nidra? → View sleep improvement recommendations

Shop Integration Unlike most health apps that stop at "you should take supplements," AyuAware seamlessly connects insights to curated products. Low Ojas leads directly to Ojas-building supplements.

3. User Journey Breakdown

3.1 First-Time Visitor Experience

Context: User navigates to Health section for the first time.

Step 1: Welcome & CHI Introduction

Twin Message:

"Welcome to your Health section!

Let me introduce you to CHI — your Constitutional Health Index."

Educational Card:

```
┌──────────────────────────────────────────────────┐
│ 💡 What is CHI?                                  │
│ ────────────────────────────────────────────────│
│                                                  │
│ CHI (Constitutional Health Index) is your        │
│ overall health score based on 7 key Ayurvedic    │
│ principles:                                      │
│                                                  │
│ • Prakriti (Your constitution)                   │
│ • Vikriti (Current state)                        │
│ • Agni (Digestive fire)                          │
│ • Ojas (Vitality essence)                        │
│ • Manas (Mental state)                           │
│ • Nidra (Sleep quality)                          │
│ • Prana (Life force)                             │
│                                                  │
│ Unlike a static score, CHI changes with your     │
│ lifestyle, reflecting your body's dynamic        │
│ state.                                           │
│                                                  │
│              [Got it! →]                         │
└──────────────────────────────────────────────────┘
```

Step 2: CHI Reveal

After user taps "Got it!":

Twin Message:

"Perfect! Let's look at your CHI score today..."

CHI Card appears with animated score fill

Why This Matters:

- User understands what they're looking at
- Sanskrit terms are introduced with English translations
- Expectation set that this is dynamic, not static
- Investment in education increases engagement

3.2 Returning User Experience

Context: User who has visited before returns to Health section.

Twin Message:

"Let's check your health today!

Your CHI score is looking good. Here's the breakdown:"

CHI Card immediately visible (collapsed)

No educational interruption — straight to value.

3.3 Exploring CHI Details

Context: User taps "See Full Analysis" on CHI card.

Expanded CHI Card Shows:

1. **Score Visualization**
    - Large circular chart with animated fill
    - Score percentage in center
    - Range label below (Excellent/Good/Fair/Needs Attention)
2. **Comparison**
    - "↑ 4% from yesterday"
    - 7-day trend line chart
    - 30-day average
3. **Full Analysis Text**
    - What's contributing positively
    - Areas for improvement
    - Specific observations from data
4. **Actions**
    - [Create CHI-Boosting Protocol →]
    - [View Detailed History →]

Example Analysis:

"Your CHI reflects steady improvement this week. Good recovery patterns in your HRV indicate strong resilience. However, a slight Kapha imbalance is showing up in:

• Lower than usual activity levels • Mild morning sluggishness • Preference for heavy, oily foods

Contributing Positively: ✓ Consistent sleep schedule ✓ Morning meditation practice ✓ Balanced meal timing

Areas for Improvement: • Increase morning movement • Lighter breakfast choices • More energizing activities"

3.4 Exploring the Seven Pillars

Context: User scrolls past CHI to see the pillars grid.

Twin Message:

"Your CHI score is built from 7 key pillars of health. Let me show you how each one is doing:"

Pillars Grid Card:

```
┌──────────────────────────────────────────────┐
│ 🏛️ The 7 Pillars of Health                    │
│ ──────────────────────────────────────────────│
│                                              │
│ Tap any pillar to explore details            │
│                                              │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐    │
│  │ Prakriti │  │ Vikriti  │  │   Agni   │    │
│  │    🌿    │  │    🔄    │  │    🔥    │    │
│  │   85%    │  │   72%    │  │   68%    │    │
│  └──────────┘  └──────────┘  └──────────┘    │
│                                              │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐    │
│  │   Ojas   │  │  Manas   │  │  Nidra   │    │
│  │    ✨    │  │    🧠    │  │    🌙    │    │
│  │   74%    │  │   81%    │  │   77%    │    │
│  └──────────┘  └──────────┘  └──────────┘    │
│                                              │
│  ┌──────────┐                                │
│  │  Prana   │                                │
│  │    💨    │                                │
│  │   82%    │                                │
│  └──────────┘                                │
│                                              │
│ ⚠️ Agni is lower than usual (68%)             │
```

```
│ This affects digestion and energy                    │
└──────────────────────────────────────────────────────┘
```

Alert Logic:

- If any pillar is below 70% → Show alert for lowest pillar
- Alert explains impact briefly
- Encourages exploration

3.5 Deep Dive into a Pillar (Agni Example)

Context: User taps the Agni pillar from the grid.

Twin Message:

"Let's look at your Agni — your digestive fire. This is the key to transforming food into energy."

Collapsed Agni Card:

```
┌──────────────────────────────────────────────────────┐
│ 🔥 Agni (Digestive Fire)                             │
│ ──────────────────────────────────────────────────── │
│                                                      │
│        [Circular Chart]                              │
│            68%                                       │
│         Fair Range                                   │
│                                                      │
│ ↓ 8% from your optimal (76%)                         │
│ ──────────────────────────────────────────────────── │
│                                                      │
│ Current State:                                       │
│ Manda Agni (Slow Digestion)                          │
│                                                      │
│ Key Signs:                                           │
│ • Afternoon sluggishness                             │
│ • Bloating after meals                               │
│ • Sweet cravings                                     │
│                                                      │
│           [See Full Details ↓]                       │
└──────────────────────────────────────────────────────┘
```

User Taps "See Full Details":

Expanded Agni Card Contains:

1. **Score + Comparison**
2. **Current State Label** — "Manda Agni (Slow Digestion)"
3. **What This Means** — Educational explanation
4. **Symptoms Observed** — With frequency data
5. **Contributing Factors** — What's causing the issue
6. **Daily Agni Curve** — Chart showing strength throughout day

7. **Insight** — "Your Agni peaks at noon, making lunch your best time for the largest meal"
8. **How to Improve** — Actionable recommendations
9. **Foods to Favor/Avoid**
10. **Actions:**
    - [Create Agni-Boosting Protocol →]
    - [View Agni History →]
    - [Shop Agni Supplements →]

3.6 Creating a Protocol from Health Insights

Context: User taps "Create Agni-Boosting Protocol" from expanded Agni card.

Twin Message:

"Great! Let me create a personalized protocol to boost your Agni.

This will include: • Nutrition recommendations • Supplement suggestions • Lifestyle practices"

Protocol Preview Card:

```
┌──────────────────────────────────────────────────┐
│ 🔥 Agni-Boosting Protocol                        │
│ ────────────────────────────────────────────────│
│                                                  │
│ 🍽️ Nutrition                                     │
│ ────────────────────────────────────────────────│
│ • Largest meal at lunch (12-1pm)                 │
│ • Start meals with ginger tea                    │
│ • Favor warm, cooked foods                       │
│ • Include digestive spices                       │
│                                                  │
│ [View Detailed Meal Plan →]                      │
│                                                  │
│ 💊 Supplements                                   │
│ ────────────────────────────────────────────────│
│ • Triphala (1 cap before bed)                    │
│ • Digestive Enzymes (with meals)                 │
│ • Ginger capsules (if needed)                    │
│                                                  │
│ [Shop Supplements →]                             │
│                                                  │
│ 🧘 Lifestyle                                     │
│ ────────────────────────────────────────────────│
│ • 10-min walk after meals                        │
│ • Eat in calm environment                        │
│ • Practice mindful eating                        │
```

```
Timeline: 2-4 weeks
Difficulty: Moderate

[Save to My Protocols]  [Start Now →]
```

Flow Continues:

- "Save to My Protocols" → Adds to user's protocol library
- "Start Now" → Activates protocol, adds tracking to Today section
- "Shop Supplements" → Opens shop filtered to Agni products

3.7 Shopping from Health Insights

Context: User taps "Shop Agni Supplements" from protocol or pillar card.

Twin Message:

"Here are supplements that can help boost your Agni:"

Product Cards:

```
🛒 Recommended for Agni

[Product Image]

Organic Triphala
90 capsules

⭐⭐⭐⭐⭐ 4.8 (127 reviews)

$24.99              [Add to Cart]

[Product Image]

Digestive Enzyme Complex
60 capsules

⭐⭐⭐⭐☆ 4.6 (89 reviews)

$29.99              [Add to Cart]
```

```
│  └──────────────────────────────────────────────────────┘│
│                                                          │
│  [View All Supplements →]                                │
└──────────────────────────────────────────────────────────┘
```

Why This Works:

- Products are contextually relevant
- User understands why these products help
- Seamless transition from insight to action
- Trust built through education first

3.8 Additional Scores Exploration

Context: User scrolls past pillars to additional scores.

Twin Message:

"Beyond the 7 pillars, I also track how stable your health patterns are over time."

Three Additional Score Cards:

1. **Dynamic Fluctuation Score (15/100)**
    - Lower is better
    - Measures consistency/stability
    - "Very Stable" — consistent lifestyle habits
2. **Metabolic Intensity Score (78/100)**
    - Higher is better
    - Measures metabolic efficiency
    - "Strong Metabolism" — efficient energy production
3. **System Energy Score (73/100)**
    - Higher is better
    - Combines Ojas + Prana + daily energy
    - "Good Energy" — solid reserves

Each card expandable with detailed breakdown.

4. Technical Architecture

4.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                              │
│                                          PRESENTATION LAYER                                  │
│                                                                                              │
│                                                                                              │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
```

```
│   │                         HealthScreen
│   │
│   │  ┌──────────┐  ┌──────────┐  ┌──────────┐
┌──────────┐   │   │
│   │  │  Header  │ │    CHI   │ │  Pillars │ │  Additional
│   │  │
│   │  │  Bar     │ │   Card   │ │   Grid   │ │   Scores
│   │  │
│   │  └──────────┘  └──────────┘  └──────────┘
└──────────┘   │   │
│
└──────────────────────────────────────────────────────────────
─┘   │
└──────────────────────────────────────────────────────────────
──────────┘
                                   │
                                   ▼
┌──────────────────────────────────────────────────────────────
──────────┐
│                          STATE LAYER
│
│   ┌──────────────────┐   ┌──────────────────┐
┌──────────────────┐
│   │   CHI Store      │   │  Pillars Store   │   │  Health UI
│   │                  │   │                  │   │  Store
│   │ • Overall score  │   │ • Pillar scores  │   │ • Expanded
cards │   │
│   │ • Trend data     │   │ • Pillar details │   │ • First
visit    │   │
│   │ • History        │   │ • Symptoms       │   │ • Scroll pos
│   │
│   │ • Contributors   │   │ • Recommendations│   │ • Active
pillar  │   │
│   └──────────────────┘   └──────────────────┘
└──────────────────┘   │
└──────────────────────────────────────────────────────────────
──────────┘
                                   │
                                   ▼
┌──────────────────────────────────────────────────────────────
──────────┐
│                          SERVICE LAYER
│
```

```
│   ┌──────────────────────┐
┌──────────────────────────────────────┐    │
│   │    Health Agent      │  │      Health Data Service
│   │                      │  │
│   │    (Claude API)      │  │
│   │                      │  │
│   │                      │  │  • CHI calculation engine
│   │                      │  │
│   │  • Introductions     │  │  • Pillar aggregation
│   │                      │  │
│   │  • Explanations      │  │  • Symptom correlation
│   │                      │  │
│   │  • Recommendations   │  │  • Protocol generation
│   │                      │  │
│   │  • Protocol creation │  │  • History tracking
│   │                      │  │
│   └──────────────────────┘
└──────────────────────────────────────┘    │
└──────────────────────────────────────────────────────────────────────────────
────────┘
                                            │
                                            ▼
┌──────────────────────────────────────────────────────────────────────────────
────────┐
│                                DATA LAYER
│
│   ┌──────────────────────┐
┌──────────────────────────────────────┐    │
│   │   Local Cache        │  │      Backend APIs
│   │                      │  │
│   │                      │  │
│   │                      │  │
│   │  • CHI data          │  │  • CHI calculation service
│   │                      │  │
│   │  • Pillar scores     │  │  • Pillar analysis service
│   │                      │  │
│   │  • Visit history     │  │  • Protocol service
│   │                      │  │
│   │  • Preferences       │  │  • Shop service
│   │                      │  │
│   └──────────────────────┘
└──────────────────────────────────────┘    │
└──────────────────────────────────────────────────────────────────────────────
────────┘
```

4.2 Data Flow: Opening Health Section

```
User navigates to Health
        |
        ▼
 ┌──────────────────────┐
 │ Check first visit    │
 └──────────────────────┘
        |
   ┌────┴────┐
   |         |
First     Returning
Visit      User
   |         |
   ▼         |
 ┌──────────────────────┐
 │ Show educational     │
 │ intro flow           │
 └──────────────────────┘
   |         |
   └────┬────┘
        |
        ▼
 ┌──────────────────────┐
 │ Load cached CHI data │ ── Instant display
 └──────────────────────┘
        |
        ▼
 ┌──────────────────────┐      Parallel requests:
 │ Fetch fresh data     │ ── • GET /health/chi
 └──────────────────────┘      • GET /health/pillars
        |                      • GET /health/scores
        ▼
 ┌──────────────────────┐
 │ Generate Twin        │ ── Claude creates contextual
 │ greeting             │    message based on data
 └──────────────────────┘
        |
        ▼
 ┌──────────────────────┐
 │ Render CHI card      │ ── Animated score fill
 │ + Pillars grid       │
 └──────────────────────┘
```

4.3 Data Flow: Pillar Expansion

```
User taps pillar in grid
        |
        ▼
```

```
┌──────────────────────┐
│ Generate Twin        │ ── Claude creates pillar
│ introduction         │    introduction message
└──────────────────────┘
          │
          ▼
┌──────────────────────┐
│ Check cached detail  │
└──────────────────────┘
          │
     ┌────┴────┐
Cached     Not Cached
     │         │
     │         ▼
     │   ┌──────────────────────┐
     │   │ Fetch pillar detail  │
     │   │ GET /health/pillars/ │
     │   │ {pillarName}         │
     │   └──────────────────────┘
     │         │
     └────┬────┘
          │
          ▼
┌──────────────────────┐
│ Show collapsed card  │
└──────────────────────┘
          │
          ▼ (user taps expand)
┌──────────────────────┐
│ Animate expansion    │ ── Height: 400ms
│ Render full detail   │    Content fade: 300ms
└──────────────────────┘
```

4.4 Data Flow: Protocol Creation

```
User taps "Create [Pillar]-Boosting Protocol"
          │
          ▼
┌──────────────────────┐
│ Generate Twin        │ ── Claude explains what
│ introduction         │    protocol will include
└──────────────────────┘
          │
          ▼
┌──────────────────────┐
│ Call protocol        │ ── POST /health/protocols/create
```

```
┌──────────────────────────┐
│ generation service       │      { pillar, userContext }
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Generate personalized    │ ── Claude creates tailored
│ protocol content         │     recommendations
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Display protocol         │ ── Nutrition + Supplements
│ preview card             │     + Lifestyle sections
└──────────────────────────┘
        │
        ├── [Save to My Protocols] ──► POST /protocols/save
        │
        ├── [Start Now] ──► POST /protocols/activate
        │
        └── [Shop Supplements] ──► Navigate to filtered shop
```

5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Health Section

Claude serves multiple functions in the Health section:

Capability	Description
Educational Introduction	Explain CHI and pillars to first-time visitors
Pillar Introductions	Contextualize each pillar before showing data
Analysis Generation	Create personalized insights from score data
Symptom Correlation	Connect user's data to Ayurvedic symptoms
Recommendation Creation	Generate actionable improvement suggestions
Protocol Generation	Create complete multi-faceted protocols
Product Contextualization	Explain why specific supplements help

5.2 System Prompt for Health Section

```
You are the Ayurvedic Digital Twin for AyuAware, serving as a
guide
through the Health section. Your role is to help users
understand their
```

Constitutional Health Index (CHI) and the 7 Ayurvedic pillars of health.

YOUR ROLE IN THIS SECTION

You are a health educator and interpreter. You don't just show numbers —
you explain what they mean, why they matter, and what to do about them.
Every metric connects to ancient wisdom and modern action.

THE 7 PILLARS YOU EXPLAIN

1. PRAKRITI (Constitution)
   - The user's birth constitution
   - Their natural dosha balance
   - Strengths and vulnerabilities

2. VIKRITI (Current State)
   - Current dosha balance vs. baseline
   - Imbalances and their causes
   - How to return to balance

3. AGNI (Digestive Fire)
   - Metabolic and digestive strength
   - Types: Sama (balanced), Vishama (irregular), Tikshna (sharp), Manda (slow)
   - Central to all health in Ayurveda

4. OJAS (Vitality)
   - Deep reserves of immunity and strength
   - Built through proper living
   - Depleted by stress, overwork, poor diet

5. MANAS (Mind)
   - Mental clarity and emotional balance
   - Three gunas: Sattva, Rajas, Tamas
   - Connection between mind and body

6. NIDRA (Sleep)
   - Quality over quantity
   - Timing relative to doshas
   - Sleep's role in healing

7. PRANA (Life Force)

- Vital energy through breath
- HRV as modern Prana indicator
- Pranayama practices

COMMUNICATION APPROACH

When introducing CHI:
- Explain it's a synthesis of 7 pillars
- Note it's dynamic, not static
- Emphasize it's personalized to their constitution

When introducing a pillar:
- Start with the Sanskrit term and meaning
- Explain its role in overall health
- Connect to their specific data

When explaining low scores:
- Frame as opportunity, not failure
- Identify likely causes from their data
- Provide actionable next steps

When recommending protocols:
- Be specific to their constitution
- Include nutrition, lifestyle, and supplements
- Set realistic timelines

LANGUAGE GUIDELINES

DO:
- Use Sanskrit terms with English translations
- Connect modern metrics to traditional concepts
- Provide specific, actionable recommendations
- Celebrate strengths while addressing weaknesses

DO NOT:
- Diagnose medical conditions
- Make alarming statements about low scores
- Recommend stopping medications
- Overwhelm with too much information at once

SCORE INTERPRETATION

85-100% (Excellent): "Thriving" — celebrate and maintain
70-84% (Good): "Solid foundation" — minor optimizations
55-69% (Fair): "Attention needed" — specific focus areas

```
<55% (Needs Attention): "Priority focus" — concentrated
effort

Always frame lower scores with hope and actionable guidance.
```

5.3 Context Building for Health Section

Every Claude call includes:

```
CONTEXT PACKAGE STRUCTURE:

## User Profile
- Name: {user_name}
- Prakriti: {prakriti_percentages}
- Constitution type: {constitution_type}
- Days active: {days_active}

## Current Health Data
- CHI Score: {chi_score}%
- CHI Trend: {chi_trend}
- CHI Contributors (positive): {positive_factors}
- CHI Contributors (negative): {negative_factors}

## Pillar Scores
- Prakriti: {prakriti_score}%
- Vikriti: {vikriti_score}%
- Agni: {agni_score}%
- Ojas: {ojas_score}%
- Manas: {manas_score}%
- Nidra: {nidra_score}%
- Prana: {prana_score}%

## Lowest Pillar Detail
- Pillar: {lowest_pillar}
- Score: {lowest_score}%
- Symptoms observed: {symptoms}
- Contributing factors: {factors}

## Recent Activity
- Check-ins this week: {checkin_count}
- Protocols active: {active_protocols}
- Last Health section visit: {last_visit}

## Visit Context
- First visit: {is_first_visit}
- Current view: {current_view}
```

```
- Pillar being explored: {active_pillar}
```

5.4 Prompt Templates by Scenario

First-Time Welcome:

```
Generate a welcoming introduction to the Health section for a
first-time visitor.

{CONTEXT_PACKAGE}

Your message should:
1. Welcome them warmly
2. Introduce CHI as their Constitutional Health Index
3. Build curiosity without overwhelming
4. Transition naturally to the educational card

Length: 2-3 short sentences
Tone: Warm, inviting, educational
```

CHI Score Insight (Returning User):

```
Generate a greeting and CHI insight for a returning user.

{CONTEXT_PACKAGE}

Your message should:
1. Acknowledge their return briefly
2. Comment on their CHI status (not just the number)
3. Hint at what's most notable (positive or needs attention)
4. Transition to showing the breakdown

Adapt tone to score:
- High score (85+): Celebratory, encouraging
- Good score (70-84): Positive, supportive
- Fair score (55-69): Constructive, actionable
- Low score (<55): Caring, reassuring, motivating

Length: 2-3 sentences
Don't repeat the exact number — that's shown in the card.
```

Pillar Introduction:

```
Generate an introduction for the {pillar_name} pillar.

{CONTEXT_PACKAGE}

Pillar Data:
- Name: {pillar_name}
```

```
- Sanskrit: {sanskrit_name}
- Score: {pillar_score}%
- Status: {status}
- Key observations: {observations}

Your introduction should:
1. Name the pillar in Sanskrit with translation
2. Briefly explain its role in health (1 sentence)
3. Transition to showing the card

Length: 2-3 sentences
Don't include detailed analysis — save that for the card.
```

Pillar Analysis (Expanded Card):

```
Generate detailed analysis for the {pillar_name} pillar.

{CONTEXT_PACKAGE}

Detailed Pillar Data:
- Name: {pillar_name}
- Score: {pillar_score}%
- Optimal for this user: {optimal_score}%
- Current state label: {state_label}
- Symptoms observed: {symptoms_with_frequency}
- Contributing factors: {factors}
- Daily pattern: {daily_pattern}

Generate analysis sections:

1. "What This Means" — Educational explanation of current
state
2. "Symptoms I've Noticed" — List with frequency data
3. "Contributing Factors" — What's causing current state
4. "Insight" — One key observation connecting data
5. "How to Improve" — 5-6 specific recommendations

Style: Educational but accessible
Include: Ayurvedic context where natural
Avoid: Medical terminology, alarming language
```

Protocol Generation:

```
Generate a personalized {pillar_name}-boosting protocol.

{CONTEXT_PACKAGE}

Current State:
```

```
- {pillar_name} score: {pillar_score}%
- Key issues: {issues}
- User's constitution: {prakriti}
- Current lifestyle factors: {lifestyle_factors}

Generate a complete protocol with:

1. NUTRITION (4-5 recommendations)
   - Specific to their constitution
   - Foods to favor and avoid
   - Meal timing suggestions

2. SUPPLEMENTS (2-3 recommendations)
   - Appropriate for the pillar
   - Include dosage guidance
   - Note any constitution-specific considerations

3. LIFESTYLE (4-5 practices)
   - Daily routines
   - Movement/exercise
   - Mind-body practices

4. METADATA
   - Timeline: How long to follow
   - Difficulty: Easy/Moderate/Challenging
   - Expected improvements

Format as structured content for card display.
```

Product Contextualization:

```
Explain why these supplements help with {pillar_name}.

{CONTEXT_PACKAGE}

Products to contextualize:
{products_list}

For each product, provide:
1. One sentence on how it helps this specific pillar
2. Connection to the user's current state
3. Any constitution-specific notes

Keep it brief — these appear as product descriptions.
```

5.5 Agni Types Explanation Framework

When explaining Agni, Claude uses this framework:

Agni Type	Sanskrit	Characteristics	Signs	Associated Dosha
Sama	समाग्नि	Balanced	Even digestion, good energy	Balanced
Vishama	विषमाग्नि	Irregular	Variable appetite, gas, bloating	Vata
Tikshna	तीक्ष्णाग्नि	Sharp/Intense	Intense hunger, acid reflux	Pitta
Manda	मन्दाग्नि	Slow/Dull	Sluggish digestion, heaviness	Kapha

5.6 Guna Framework for Manas

When explaining Manas (Mind), Claude uses the three gunas:

Guna	Sanskrit	Quality	Manifestation
Sattva	सत्त्व	Clarity, harmony	Clear thinking, emotional stability, wisdom
Rajas	रजस	Activity, passion	Restlessness, ambition, agitation
Tamas	तमस	Inertia, dullness	Lethargy, confusion, resistance

Ideal balance: High Sattva, moderate Rajas, low Tamas

6. Data Models & State Management

6.1 Core Data Types

CHI Score:

```
CHIScore:
  - value: number (0-100)
  - status: 'excellent' | 'good' | 'fair' | 'needs_attention'
  - trend: 'improving' | 'stable' | 'declining'
  - changeFromYesterday: number (can be negative)
  - changeFromWeekAgo: number
  - sevenDayHistory: DailyScore[]
  - thirtyDayAverage: number
  - contributors: CHIContributor[]
  - lastCalculated: Date

DailyScore:
  - date: Date
  - value: number

CHIContributor:
```

```
  - factor: string
  - impact: 'positive' | 'negative'
  - weight: number
  - description: string
```

Pillar Data:

```
Pillar:
  - id: PillarType
  - name: string
  - sanskritName: string
  - icon: string (emoji)
  - color: string (hex)
  - score: number (0-100)
  - status: 'excellent' | 'good' | 'fair' | 'needs_attention'
  - optimalForUser: number
  - lastUpdated: Date

PillarType: 'prakriti' | 'vikriti' | 'agni' | 'ojas' |
'manas' | 'nidra' | 'prana'

PillarDetail:
  - pillar: Pillar
  - stateLabel: string (e.g., "Manda Agni")
  - stateLabelSanskrit: string
  - explanation: string
  - symptoms: Symptom[]
  - contributingFactors: Factor[]
  - dailyPattern: DailyPattern (optional)
  - recommendations: Recommendation[]
  - history: DailyScore[]

Symptom:
  - description: string
  - frequency: string (e.g., "7×/week")
  - severity: 'mild' | 'moderate' | 'significant'

Factor:
  - description: string
  - isModifiable: boolean
  - suggestedAction: string

DailyPattern:
  - times: string[] (e.g., ["6AM", "12PM", "8PM"])
  - values: number[]
  - peakTime: string
```

```
  - insight: string

Recommendation:
  - category: 'nutrition' | 'lifestyle' | 'practice' |
'supplement'
  - action: string
  - timing: string (optional)
  - priority: 'high' | 'medium' | 'low'
```

Additional Scores:

```
AdditionalScore:
  - type: 'dynamic_fluctuation' | 'metabolic_intensity' |
'system_energy'
  - name: string
  - value: number (0-100)
  - interpretation: string
  - icon: string
  - isLowerBetter: boolean (true for fluctuation)
  - details: ScoreDetail[]

ScoreDetail:
  - metric: string
  - value: string
  - trend: 'up' | 'down' | 'stable'
```

Protocol:

```
Protocol:
  - id: string
  - name: string
  - targetPillar: PillarType
  - nutrition: NutritionPlan
  - supplements: Supplement[]
  - lifestyle: LifestylePractice[]
  - timeline: string
  - difficulty: 'easy' | 'moderate' | 'challenging'
  - expectedOutcomes: string[]
  - createdAt: Date
  - isActive: boolean

NutritionPlan:
  - guidelines: string[]
  - foodsToFavor: string[]
  - foodsToAvoid: string[]
  - mealTimingSuggestions: string[]

Supplement:
```

```
  - productId: string
  - name: string
  - dosage: string
  - timing: string
  - notes: string

LifestylePractice:
  - practice: string
  - frequency: string
  - duration: string (optional)
  - instructions: string (optional)
```

6.2 State Structure

```
HealthState:
  # Visit Context
  - isFirstVisit: boolean
  - lastVisit: Date | null
  - hasSeenEducation: boolean

  # CHI Data
  - chiScore: CHIScore | null
  - isLoadingCHI: boolean
  - chiError: Error | null

  # Pillars Data
  - pillars: Pillar[]
  - pillarDetails: Record<PillarType, PillarDetail | null>
  - activePillar: PillarType | null
  - isLoadingPillar: boolean

  # Additional Scores
  - additionalScores: AdditionalScore[]
  - isLoadingScores: boolean

  # Protocols
  - generatedProtocol: Protocol | null
  - isGeneratingProtocol: boolean

  # UI State
  - expandedCards: string[]
  - scrollPosition: number
  - showEducationModal: boolean

  # Chat
  - messages: HealthMessage[]
```

```
  - isTyping: boolean
```

6.3 State Management Approach

We use **Zustand** with **separate stores**:

```
Stores:

1. chiStore
   - CHI score and history
   - Contributors
   - Trend calculations
   - Refresh logic

2. pillarsStore
   - All 7 pillar scores
   - Pillar detail data (lazy loaded)
   - Symptom correlations
   - Recommendation generation

3. healthUIStore
   - First visit tracking
   - Expanded cards
   - Active pillar
   - Scroll position
   - Messages

4. protocolStore
   - Generated protocols
   - Protocol activation
   - Progress tracking
```

7. Component Requirements

7.1 CHI Overview Components

CHICard

Displays the main CHI score with:

- Animated circular chart
- Score percentage
- Status label
- Trend indicator
- Quick insight text
- Expand button

States:

- Collapsed: Score, trend, quick insight
- Expanded: Full analysis, history chart, actions

CHICircularChart

- Donut-style visualization
- 180x180px size
- 20px stroke width
- Animated fill (800ms)
- Color based on score range
- Score displayed in center

CHIHistoryChart

- Line chart showing 7-day trend
- 120px height
- Teal gradient fill
- Data points marked
- Averages displayed below

7.2 Pillar Components

PillarsGridCard

Container showing all 7 pillars:

- 3-column grid (mobile)
- 4-column grid (desktop)
- Tap to select pillar
- Alert section for lowest pillar

PillarItem

Individual pillar in grid:

- 100x100px
- Icon centered top
- Name below icon
- Score below name
- Color-coded border
- Tap state animation

PillarDetailCard

Full pillar deep-dive:

- Circular score chart
- State label (Sanskrit + English)
- Expandable sections
- Symptoms with frequency
- Contributing factors
- Daily pattern chart (if applicable)
- Recommendations
- Action buttons

7.3 Pillar-Specific Components

AgniDailyCurve — Line chart showing Agni strength by time of day

DoshaComparisonChart — For Prakriti/Vikriti showing baseline vs current

GunaBalanceBar — Horizontal bar showing Sattva/Rajas/Tamas distribution

SleepPatternChart — For Nidra showing sleep phases and timing

PranaHRVChart — For Prana showing HRV and breathing patterns

7.4 Additional Score Components

AdditionalScoreCard

For Dynamic Fluctuation, Metabolic Intensity, System Energy:

- Custom icon visualization
- Score value
- Interpretation label
- Brief explanation
- Expandable detail

7.5 Protocol Components

ProtocolPreviewCard

Shows generated protocol:

- Three sections: Nutrition, Supplements, Lifestyle
- Expandable within each section
- Product cards for supplements
- Timeline and difficulty indicators
- Save and Start buttons

ProductCard

Shop product display:

- Product image (120x120px)
- Name and size
- Star rating
- Price
- Add to Cart button

7.6 Educational Components

EducationalCard

First-time education:

- Light teal background (#F0F9F7)
- 💡 icon
- Bulleted content
- "Got it!" button
- Dismissible

PillarEducationTooltip

Quick explanation of any pillar term:

- Triggered by info icon tap
- Sanskrit term + translation
- 1-2 sentence explanation
- Dismissible

7.7 Animation Requirements

Animation	Duration	Easing	Trigger
CHI chart fill	800ms	ease-out	Card appears
Pillar item tap	100ms	ease-out	Touch
Card expand	400ms	ease-out	Expand button
Content fade-in	300ms	ease-in	After expand
Score color pulse	200ms	ease-in-out	Score change
History chart draw	600ms	ease-out	Card expands
Alert appearance	300ms	ease-out	Data load

8. API Requirements

8.1 REST Endpoints

Get CHI Overview:

```
GET /api/v1/health/chi

Response:
{
  "score": {
    "value": 72,
    "status": "good",
    "trend": "improving",
    "changeFromYesterday": 4,
    "changeFromWeekAgo": 8
  },
  "history": [
    { "date": "2025-01-25", "value": 68 },
    { "date": "2025-01-26", "value": 70 },
    ...
  ],
  "contributors": {
    "positive": [...],
```

```
    "negative": [...]
  },
  "lastCalculated": "2025-01-31T08:00:00Z"
}
```

Get All Pillars:

```
GET /api/v1/health/pillars

Response:
{
  "pillars": [
    {
      "id": "prakriti",
      "name": "Prakriti",
      "sanskritName": "प्रकृति",
      "icon": "🌿",
      "color": "#A3B899",
      "score": 85,
      "status": "excellent",
      "optimalForUser": 85
    },
    ...
  ],
  "lowestPillar": {
    "id": "agni",
    "score": 68,
    "alertMessage": "Agni is lower than usual..."
  }
}
```

Get Pillar Detail:

```
GET /api/v1/health/pillars/{pillarId}

Response:
{
  "pillar": { ... },
  "stateLabel": "Manda Agni",
  "stateLabelSanskrit": "मन्दाग्नि",
  "explanation": "Your digestive fire is dampened...",
  "symptoms": [
    { "description": "Afternoon energy crashes", "frequency":
"7×/week", "severity": "moderate" }
  ],
  "contributingFactors": [...],
  "dailyPattern": {
```

```
    "times": ["6AM", "12PM", "8PM"],
    "values": [45, 85, 52],
    "peakTime": "12PM",
    "insight": "Your Agni peaks at noon..."
  },
  "recommendations": [...],
  "history": [...]
}
```

Get Additional Scores:

```
GET /api/v1/health/scores

Response:
{
  "scores": [
    {
      "type": "dynamic_fluctuation",
      "name": "Dynamic Fluctuation Score",
      "value": 15,
      "interpretation": "Very Stable",
      "isLowerBetter": true,
      "details": [...]
    },
    ...
  ]
}
```

Create Protocol:

```
POST /api/v1/health/protocols/create

Request:
{ "pillarId": "agni", "focus": "boost" }

Response:
{
  "protocol": {
    "id": "proto_xxx",
    "name": "Agni-Boosting Protocol",
    "targetPillar": "agni",
    "nutrition": { ... },
    "supplements": [ ... ],
    "lifestyle": [ ... ],
    "timeline": "2-4 weeks",
    "difficulty": "moderate"
  },
  "recommendedProducts": [...]
```

```
}
```
Save/Activate Protocol:

```
POST /api/v1/protocols/save
POST /api/v1/protocols/activate
```
Get Pillar History:

```
GET /api/v1/health/pillars/{pillarId}/history?days=30
```
Get Supplements for Pillar:

```
GET /api/v1/shop/supplements?pillar={pillarId}
```
8.2 Deep Linking

Path	Destination
`/health`	CHI overview
`/health/chi`	CHI detail (expanded)
`/health/prakriti`	Prakriti pillar
`/health/vikriti`	Vikriti pillar
`/health/agni`	Agni pillar
`/health/ojas`	Ojas pillar
`/health/manas`	Manas pillar
`/health/nidra`	Nidra pillar
`/health/prana`	Prana pillar

9. Success Metrics

9.1 Engagement Metrics

Metric	Target	Measurement
Section visits per week	> 3	Users visiting Health weekly
Pillar exploration rate	> 50%	Users who tap at least 3 pillars
Card expansion rate	> 60%	Expanded cards / shown cards
Time in section	> 3 minutes	Average session duration
Return within 7 days	> 60%	Users who revisit Health

9.2 Education Metrics

Metric	Target	Measurement
Education completion	> 90%	First-time users who complete intro
Pillar comprehension	Survey	Post-onboarding quiz (optional)

9.3 Action Metrics

Metric	Target	Measurement
Protocol creation rate	> 20%	Users who create at least 1 protocol
Protocol activation rate	> 50%	Created protocols that are activated
Shop click-through	> 15%	Users who tap "Shop Supplements"
Add to cart from Health	Track	Products added from Health section

9.4 Health Outcome Metrics

Metric	Target	Measurement
CHI improvement	Positive trend	Week-over-week CHI change
Pillar improvement	Track	Individual pillar score changes
Protocol completion	> 40%	Users who complete activated protocols

9.5 Technical Metrics

Metric	Target	Measurement
Section load time	< 2s	Time to first meaningful content
CHI calculation time	< 1s	Backend calculation duration
Pillar expansion time	< 500ms	Time from tap to expanded
Protocol generation time	< 3s	Time to generate personalized protocol
Error rate	< 1%	Failed loads or calculations

10. Implementation Considerations

10.1 Performance Optimization

CHI Calculation:

- Calculate server-side on schedule (hourly)
- Cache result, serve from cache
- Recalculate on-demand only if data changed significantly

Pillar Data Loading:

- Load all pillar summaries on section open
- Lazy-load pillar details on first expansion
- Cache detail data for session
- Prefetch likely-next pillar (lowest score)

Chart Rendering:

- Use SVG for circular charts
- Animate with CSS transforms (GPU-accelerated)
- Throttle history chart updates

10.2 Offline Handling

When Offline:

- Show cached CHI and pillar scores
- Display "Last updated X ago" indicator
- Allow viewing cached pillar details
- Disable protocol creation (requires Claude)
- Queue any actions for sync

When Back Online:

- Sync queued actions
- Refresh CHI score
- Update pillar data

10.3 Error Handling

CHI Calculation Failed:

- Show last known score with timestamp
- Offer manual refresh option

Pillar Data Failed:

- Show skeleton in grid
- "Tap to retry" on individual pillar

Protocol Generation Failed:

- Retry once automatically
- Show friendly message if still fails

10.4 Accessibility Requirements

Screen Reader:

- CHI score announced fully
- Pillars announced with name and score
- Charts have text alternatives
- Expand state announced

Motor Accessibility:

- All tap targets ≥ 44x44px
- Pillar items 100x100px
- No time-sensitive interactions

Visual Accessibility:

- WCAG AA contrast compliance
- Don't rely solely on color (use icons + labels)
- Support Dynamic Type

10.5 Analytics Events

Event	Properties	Purpose
`health_opened`	source, is_first_visit	Track entry
`chi_viewed`	score, status	CHI engagement
`chi_expanded`	score	Detail engagement
`pillar_tapped`	pillar_id, score	Pillar interest
`pillar_expanded`	pillar_id	Detail depth
`protocol_created`	pillar_id	Action taken
`protocol_activated`	protocol_id	Commitment
`shop_clicked`	pillar_id, source	Commerce interest
`product_added`	product_id, pillar_id	Conversion
`education_completed`	time_spent	Onboarding success

10.6 Testing Priorities

Unit Tests:

- CHI calculation logic
- Pillar score aggregation
- Status threshold logic

Integration Tests:

- Full section load flow
- Pillar expansion with data fetch
- Protocol creation end-to-end

E2E Tests:

- First-time user education flow
- Returning user CHI view → Pillar exploration
- Protocol creation → Save → Activate

Manual QA:

- CHI chart animation smoothness
- Protocol relevance to constitution
- Education clarity for new users

Appendix A: Pillar Reference Guide

Prakriti (Constitution) — 🌿

Sanskrit: प्रकृति (Prakṛti) **Meaning:** "Original creation" — your birth constitution **Color:** #A3B899 (sage green)

What It Measures: Alignment with natural dosha balance

Vikriti (Current State) — 🔄

Sanskrit: विकृति (Vikṛti) **Meaning:** "Changed state" — current vs. baseline **Color:** #E89B7F (coral)

What It Measures: Deviation from Prakriti, current imbalances

Agni (Digestive Fire) — 🔥

Sanskrit: अग्नि (Agni) **Meaning:** "Fire" — metabolic strength **Color:** #F8C471 (gold)

Agni Types: Sama (balanced), Vishama (irregular), Tikshna (sharp), Manda (slow)

Ojas (Vitality) — ✨

Sanskrit: ओजस् (Ojas) **Meaning:** "Vigor" — essence of vitality **Color:** #F0E68C (light gold)

What It Measures: Deep reserves, immunity, resilience

Manas (Mind) — 🧠

Sanskrit: मनस् (Manas) **Meaning:** "Mind" — mental state **Color:** #B8A8E1 (lavender)

Gunas: Sattva (clarity), Rajas (activity), Tamas (inertia)

Nidra (Sleep) — 🌙

Sanskrit: निद्रा (Nidrā) **Meaning:** "Sleep" — rest quality **Color:** #7AB8AC (soft teal)

Focus: Quality over quantity, timing with doshas

Prana (Life Force) — 💨

Sanskrit: प्राण (Prāṇa) **Meaning:** "Life force" — vital energy **Color:** #A8E6A3 (light green)

What It Measures: HRV, breathing patterns, vitality

Appendix B: Protocol Templates

Agni-Boosting Protocol

Nutrition: Largest meal at lunch, ginger tea, warm foods, digestive spices **Supplements:** Triphala, Digestive Enzymes, Ginger **Lifestyle:** Walk after meals, calm eating environment, mindful eating **Timeline:** 2-4 weeks | **Difficulty:** Moderate

Vata-Balancing Protocol

Nutrition: Regular meal times, warm/moist foods, healthy fats **Supplements:** Ashwagandha, Triphala, Sesame oil **Lifestyle:** Consistent routine, oil massage, warm baths, gentle yoga **Timeline:** 4-6 weeks | **Difficulty:** Moderate

Ojas-Building Protocol

Nutrition: Sattvic foods, ghee, dates, almonds, fresh meals **Supplements:** Chyawanprash, Shatavari/Ashwagandha **Lifestyle:** Adequate sleep, meditation, stress reduction **Timeline:** 4-8 weeks | **Difficulty:** Easy to Moderate

End of Document